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Fine-tuning Large Language Models

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Introduction to Fine-Tuning LLMs

What is fine-tuning?

- What are some ways in which you have been using LLMs?
  - Prompting
  - Using code generation tools
  - Creating applications using frameworks
  - Retrieval-augmented generation
  - Using embeddings generated by LLMs
- How can we customize these LLMs further to suit our specific needs? For example:
  - Performing tasks on a specialized knowledge base
  - Setting the style, tone, or format of the output
  - Handling edge cases

Examples of customization

"Tired even after sleeping, weight gain, dry skin, always cold, irregular periods, and difficulty concentrating."

Non fine-tuned LLM

Likely hypothyroidism. Recommend thyroid blood tests and endocrinologist referral.

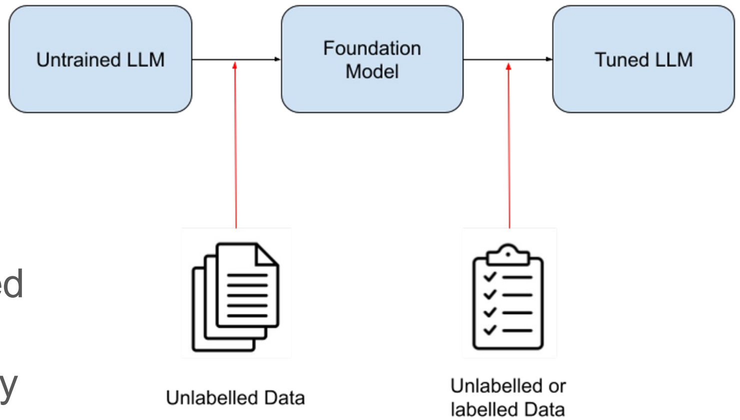
Fine-tuned LLM

Primary hypothyroidism, possibly Hashimoto's thyroiditis. Recommend anti-TPO antibody test and a full thyroid panel (TSH, Free T3, Free T4, Reverse T3). Will start levothyroxine therapy and monitor dosage.

Fine-tuning an LLM for specialized knowledge in endocrinology

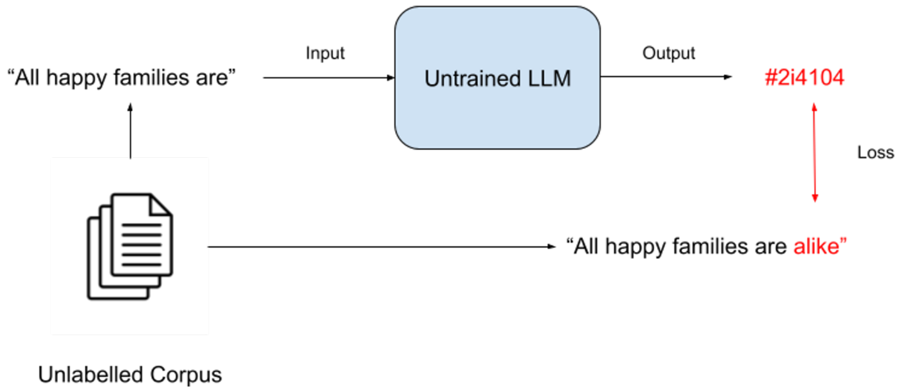
Training LLMs

- LLMs are usually trained in two stages:
  - Pre-training
  - Fine-tuning
- Pre-training gives us the foundation model:
  - Training is done on a large unlabelled corpus of data
  - Takes a lot of time and compute resources
- The foundation model received after pre-training is then fine-tuned:
  - Training is done on a smaller data set (may be labelled or unlabelled)
  - Takes less time and compute resources comparatively
- Let's examine these two stages of training LLMs in more detail



Pre-training

- The untrained LLM starts with randomized weights
- Inputs and outputs are in the form of "tokens"
- The task is next token prediction
- *Self-supervised learning*: No need to collect labels
- The model learns:
  - Language
  - General knowledge
- Expensive and time consuming



Shortcomings of Foundation Models

- The output is the next predicted token, not a “response” to your input.
- For example
  - Input: “How to tie a tie?”
  - Actual Output: “How to tie a tie? This is a question that many men ask themselves. The answer is not as simple as it seems. There are many different ways to tie a tie… ”
  - Desired output: “Tying a tie can seem like a daunting task, but with a little practice, you'll be a pro in no time! Here's a step-by-step guide on how to tie a tie:

    1. Start by facing a mirror and standing up straight.
    2. Place the tie around your neck with the wide …”
- Only contains knowledge about general topics included in the pre-training corpus
- May not be useful in cases where we need knowledge of specialized topics

Fine tuning

- Fine tuning refers to the retraining of a foundation model
  - For example GPT-3 was fine-tuned to get ChatGPT
- Training can be of both types:
  - Self-supervised learning on unlabelled data
  - Supervised learning on labelled data
- Fine-tuning can achieve one or both of the following results:
  - To increase the knowledge of the model
  - To change the behaviour of the model

Fine-Tuning

- Fully fine-tuning a base LLM updates all the weights in the model
- This may lead to catastrophic forgetting
- In case of unlabelled data, the training process looks similar to that seen in pre-training
- In case of labelled data, the training data needs to be set into a prompt before it is sent in for training
- In both cases, the difference between the actual output and the predicted output is used to calculate the loss

Training with Labelled Data

Instruction	Give three tips for staying healthy.	Describe a time when you had to make a difficult decision.	…
Response	1.Eat a balanced diet and make sure to include plenty of fruits and vegetables. 2. Exercise regularly to keep your body active and strong …	I had to make a difficult decision when I was working as a project manager at a construction company. I was in charge of a project that needed to be completed …	…

Input

```
"""Below is an instruction that
describes a task. Write a response
that appropriately completes the
request.

### Instruction:
{instruction}

### Response:"""
```

Output

```
"{response}"
```

Why Fine-tune?

- Fine-tuning can improve the model performance
  - More consistent responses
  - Fewer hallucinations
  - Better moderation
- Small fine-tuned models can outperform larger ones
  - This may lead to lower costs
  - For example, a smaller model (with 410 million parameters) fine-tuned for a specialized task may give better performance on that task than a larger non-fine-tuned model (with 175 billion parameters)
- Overcomes finite context window challenges
  - The model can only handle a specific number of tokens as input in a prompt
  - As the input size increases, the initial tokens get truncated out of the input

Fine-tuning vs. Prompt Engineering

Fine-tuning

- Large amounts of new data can be used
- Require large amounts of data (which can be expensive)
- Some upfront costs
- Some technical knowledge required
- Performance is limited by the performance of the foundation model and the data provided
- Should be tried if prompt engineering doesn’t give good results

Prompt Engineering

- Amount of new data input is limited by token size
- Don’t need new data to get started
- Little upfront costs
- Less technical knowledge required
- Performance suffers because of these limitations
- Should be the first approach while solving a problem

Approaches to Fine-Tuning

- Self-supervised learning
  - Trains on the inherent structure of the data
  - The task remains the same: next-token prediction
  - Example: it can be used to mimic writing styles
- Supervised learning
  - Trains on input-output pairs
  - Curating a training data set is key
  - Example: Q&A pairs in a prompt template
- Reinforcement learning
  - Uses a reward model to guide training
  - Aligns model responses with human preferences

Steps required in fine-tuning

1. Choose fine-tuning task
2. Prepare training data set
3. Choose a base model
4. Fine-tune via supervised learning [using code/GUI]
5. Evaluate model performance [Did it work?]

GUI Based Solutions for Fine-Tuning

https://platform.openai.com/finetune

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Fine-Tuning Code